Split-graph factor bases: exact audit and measured F4 degree

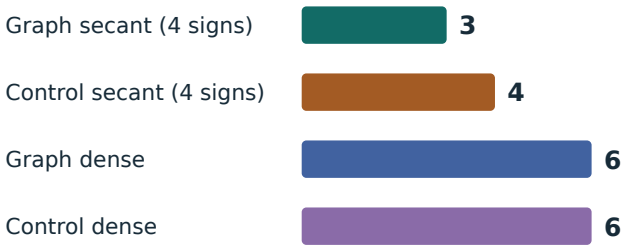
2026-10-31 · run-20261031T201044Z · two summands · prime fields

The complete toy audit has zero mismatches over 707,472 structured and 10,404 dense assignments. All nine prospective graph-construction cells accepted. The actual F4 handled degree drops, while four-branch row work exceeds the same-base dense reference. The plot reports native aggregates, with all signs included; it makes no timing or complete field-operation claim.

Split-graph bases: degree and all-branch work

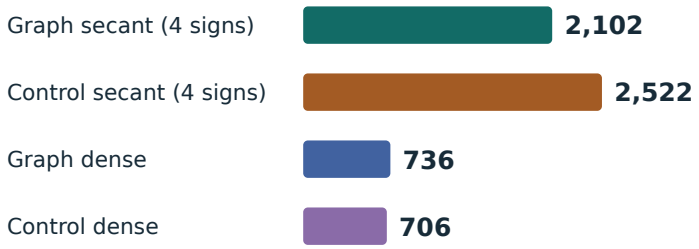
run-20261031T201044Z · stock neogb F4 · exact counts; clocks are diagnostic

A Toy handled degree



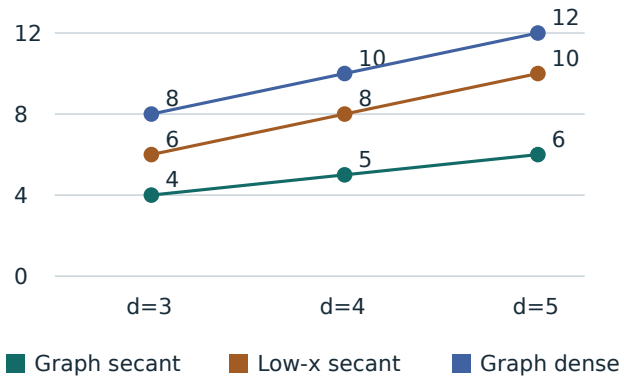
18 targets/arm; degree=max(input, selected-pair LCM).

B Toy reduced rows (all branches)



All four sign branches charged; dense has one solve.

C Larger degree comparison (F83)



D Graph-construction attempts

Field	d=3	d=4	d=5
F83	1	4	23
F101	3	10	260
F127	1	2	62

9/9 prospective cells accepted within frozen caps.

Lower degree did not lower the all-branch row proxy.

14,194/14,220 original extractions verified; 26 caps retained. Separate exact saved-basis replay: 14,220/14,220.

Source: native compact summary, results and evidence, frozen protocol. L0 shared-host execution.

Run record, comparison boundary and limits

Requested and executed: three tracks in parallel, followed by a PR: complete F17 audit; matched stock neogb F4; prospective F83/F101/F127 construction at graph degrees 3, 4 and 5. F4 consumed the larger exports after construction, retaining both low-x and hash-x controls and every affine target.

Original F4 flow	14,220 attempted; 14,194 verified; 26 root-extraction assignment caps.
Separate offline replay	14,220/14,220 saved finite-field root sets verified; zero new CAS calls. Original extraction statuses remain unchanged.
Observed handled degree	max(input-generator degree, maximum selected-pair LCM degree). Not sugar or full matrix-row degree.
Homogeneous theorem	Specified top-ideal invariant d+1 versus 2d+1 under complete squarefree splitting. Distinct from measured F4 degree.
All-branch work	Toy graph secant: 2,102 reduced rows; same-base dense: 736. Matched nongraph secant: 2,522. These are component proxies.
Natural toy yield	Graph covers18/18 finite targets; control16/18. Both have56 total finite ordered relations, eight doubles and32 unordered pairs.
Unavailable	Complete field-operation cost, independent-row yield, complete cost per usable relation and qualified timing speedup remain null.

Model: completely split graph intersection $h(T)^2-f(T)$ supplies the exact signed factor base. The secant model retains x,z,m and all four ordinate-sign branches. Doubles and target-equal points are included; the identity is handled separately by inverse pairs. Offline replay proves finite-field root-set equality by two inclusions and exact group references; it does not certify ideal equality or multiplicities, or improve relation-discovery runtime.

Execution and freeze: native Rust/C; pinned stock msolve-neogb revision `3672113ee2c696a2ac9c637636e0f7214f500ba1`, exact sparse mode2, seed315, one thread. The bounded local exception was recorded:1 CPU/2GiB per track; audit120s, ladder180s outer/120s internal, F4 1,800s combined/10s per child. No paid resources. Checkpoint admission remained failed; it was not relabelled.

Evidence: source/binary lock, archive and replay commands, post-data amendment, independent data review, mathematics review. Raw run and companion are retained in the deterministic, hash-addressed archive.

Curve identities: toy `icv1-fp5-tm1-a77cc6af`; F83 `icv1-fp7-t5-de12415d`; F101 `icv1-fp7-t5-b172aa85`; F127 `icv1-fp7-tm11-1cabae54`. Central registry/EC1 linkage and literature-wide novelty remain open.

Prior work: MIT group-law notes Petit–Kosters–Messeng, PKC2016 stock msolve. The earlier identity certificate is IDC1h621ed1a51b5f3ca8.

Next direction: share elimination across signs and implement exact triangular root extraction, then prospectively compare complete same-base relation cost. Canonical priced-pipeline dashboards receive no unvalidated cost row from this component study. The task dashboard displays the measured comparison.